Project Phase 1: Data Requirements

CS4.301: Data and Applications
Team 54: Apparatus

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Outline

Mini-world. The WCA is the worldwide body that sanctions competitions in "twisty puzzles" (Rubik's Cube and 16 other official events), publishes the regulations they run under, and keeps the official results, records and rankings. The database captures the competitive core of that world:

- competitions and their venues and dates,
- the events and rounds held at each,
- the formats, time limits, cutoffs and advancement rules of each round,
- the people who register, compete, judge, scramble, organise and delegate,
- every attempt and the results derived from it,
- scrambles, records and rankings, and
- the delegate and committee structure that governs it all, down to countries and continents.

One shared, constraint-checked store lets organizers and delegates run a competition, lets the Results Team publish and correct results once for everyone, and lets competitors, statisticians and the public see consistent records and rankings computed from millions of attempts. Its users are:

- competitors,
- organizers,
- WCA Delegates (trainee to senior),
- the WCA teams and committees that announce competitions and post results, and
- public viewers.

Later sections details how each interacts with it.

1 Domain Study & Idea Formulation

The WCA mini-world is a rule-bound pipeline. It follows the following procedure:

1. A competition is announced
2. People register for the competition
3. Rounds of the competition are run under strict regulations
4. Every attempt is recorded, and
5. Results of the competition update world-wide records and rankings.

The facts below come from the live site, its Regulations (version April 1, 2026), its public Results Export (v2.0.2, export dated October 13, 2026) and its public API.

1.1 The organisation and its history

- The WCA was formed in 2004 by Ron van Bruchem (Netherlands) and Tyson Mao (USA), after the 2003 World Championship in Toronto; the first regulations were released in August 2004 (Speedcubing History). The 1982 Budapest championship was added retroactively as a World Championship.
- Governance: a Board, WCA Delegates organised into regions, and teams and committees. Team identifiers seen in the API include `board`, `wrc` (Regulations Committee), `wcat` (Competition Announcement Team), `weat`, `wfc` (Financial), `wqac`, `wst` (Software), `wic`, `wmct` and `wsot`; each membership carries `leader` and `senior_member` flags (competition API, WC2025). The full expansion of every acronym is not shown on the pages we could read (the Teams page renders client-side).
- Delegates have a status of `trainee_delegate`, `junior_delegate`, `delegate`, `regional_delegate` or `senior_delegate`, a `region_id` and a location (WC2025 API). Trainees are recommended by a Delegate, appointed by the Senior Delegate, and cannot delegate without a Junior Delegate or higher (Delegate Handbook).

1.2 Events and round formats

Regulation 9b fixes 17 official events and the formats each may use. "Average of 5" drops the best and worst attempt and averages the other three; one DNF/DNS is allowed, two make the average DNF (9f8–9f9). "Mean of 3" is DNF if any attempt is DNF (9f11).

Retired events (`333ft`, `magic`, `mmagic`, `333mbo`) still appear in historical records and personal bests (records API), so the event list must support an official/retired status. A new Head to Head format exists for the final round of `333`, `444`, `333bf` and `333oh` only (9b7).

1.3 Competitions

A competition record (example WC2025) carries an ID (`WC2025`), name, short name, city, country, venue, street address, venue details, latitude and longitude, start and end dates (3–6 July 2025), competitor limit (2000), base entry fee in the currency's lowest denomination (12000 = USD 120), registration open and close timestamps, refund policy (50

Event IDs (as in the export)	Full round format	Cutoff phase
`333`, `222`, `444`, `555`, `333oh`, `clock`, `minx`, `pyram`, `skewb`, `sq1`	Average of 5	Best of 2
`666`, `777`	Mean of 3	Best of 1
`333bf`	Best of 5 (Ao5 also ranked)	—
`444bf`, `555bf`	Best of 3 (Mo3 also ranked)	—
`333fm`	Best of 1 or 2, or Mean of 3	Best of 1 or 2
`333mbf`	Best of 1, 2 or 3	Best of X before Best of Y, X $<$ Y

Table 1: Event formats and cutoff phases

1.4 Rounds, groups and scrambles

- An event has at most four rounds; a round with ≤ 99, ≤ 15 or ≤ 7 competitors may be followed by at most 2, 1 or 0 further rounds (9m). At least 25% must be eliminated between rounds (9p1), and advancement is by ranking or by result (9p2).
- Each round must announce its format, time limit, cutoff and advancement condition before it starts (9p2a). The default time limit is 10 minutes per attempt; a cumulative limit may span rounds (A1a). Two consecutive rounds may be Dual Rounds, where no one is eliminated (9v).
- Competitors in a round may be split into groups, each with its own scramble set (1h). Scrambles must come from the official program and stay secret until the group ends (4b). Round IDs in the WCIF look like `333-r1`.

1.5 Results, records and rankings

- Times under 10 minutes are truncated to hundredths; 10 minutes and over to seconds (9f1–9f2). The export stores times in centiseconds, DNF as -1, DNS as -2, "no result" as 0; Fewest Moves as move counts (averages $\times 100$); Multi-Blind as an encoded number in which smaller is better (Results Export README).
- Records exist at national, continental and world level; a result equal to or better than the current record counts (9i1). All results of a round are dated to the round's last local calendar day (9i2). The records API keys continental records by IDs like `_Asia` and national ones by country IDs like `India` (records API).
- A person profile shows competition count, medals and record counts, and per-event single and average bests with world, continental and national ranks; for example Max Park (2012PARK03): 218 competitions, 741 medals, 92 world records (person API).

1.6 People and officials

- A competition needs a WCA Delegate and an organization team with judges, scramblers and score takers; Head to Head adds announcers; anyone may hold several roles and also compete (Article 1).
- Competitors register with name, country, date of birth, contact details and chosen events (2c). Name, country, gender and results are public; everything else is confidential (2d).

A competitor represents a country of citizenship and may change it at most once a year (2e2).

- WCA IDs look like `2012PARK03`: year of first competition, four letters from the surname, a two-digit counter. The export's `persons.sub_id` keeps earlier name and country versions.
- Staff assignments in the WCIF use codes like `competitor`, `staff-judge`, `staff-scrambler`, `staff-runner` and `staff-Delegate`, sometimes with a station number (WC2025 WCIF).

1.7 Scope

In scope	Why
Competitions, venues (as attributes), championship designations	The unit every result hangs off of
Events, formats, rounds, cutoffs, time limits, advancement	Needed to validate and rank results (Article 9)
Registrations and selected events	Drives who may compete in which round
Results, individual attempts, scrambles	The WCA's primary data product
Records and rankings (derived)	Most-used public outputs; kept as derived data
Persons, countries, continents, name/country history	Identity and representation rules (Article 2)
Delegates, regions, supervision, organizers, teams, staff assignments	Who runs and governs competitions (Article 1)

Table 2: Scope and rationale

Out of scope	Why
Payments, refunds, donations, spectator tickets	Financial processing belongs to payment providers, not the results database
Detailed schedule activities and room layouts	Logistics data, changes minute by minute; we keep rounds and groups only
Head to Head match brackets	New format; we flag H2H rounds but do not model individual matches
Incident and disciplinary logs, delegate report text	Free-text documents, not structured results
Forum, media, translations, avatars, user authentication	Website features unrelated to competition data
Unofficial events	Excluded from official results by 9b++
Multi-location Fewest Moves simultaneous competitions	Rare special case that breaks the one-country-per-competition rule

Table 3: Out of scope items and rationale

2 Users Identification & Interaction

Eight user groups read and write different slices of the same data; each maps to an end-user category from Elmasri & Navathe Ch. 1.4.

Expected interactions:

User	**Category (Elmasri §1.4)**	**Permissions & Access Actions**
Competitor	Naive / parametric, via canned forms	**Reads:** Own registrations, results, personal bests, ranks, upcoming competitions. **Writes:** Registration (events, status changes such as withdrawal), profile change requests.
Organizer	Parametric	**Reads:** Registrations, accepted count against the limit, staff availability. **Writes:** Competition draft, schedule of rounds, registration acceptance, staff assignments.
WCA Delegate (trainee, junior, full)	Parametric plus casual queries	**Reads:** Registrations, rounds, scrambles, results in progress. **Writes:** Results and attempts, extra attempts, disqualifications, scrambles, round parameters before a round starts.
Regional / Senior Delegate	Casual	**Reads:** Delegate workload, competitions in their region. **Writes:** Delegate appointments, status changes, supervision links.
WCA Results Team	Sophisticated	**Reads:** Whole results base, duplicate-person checks. **Writes:** Posting results, corrections, name and country changes (Reg. 2c2), merges.
Competition Announcement Team	Casual	**Reads:** Competition drafts, delegate assignment, nearby competitions. **Writes:** Announcement and cancellation timestamps.
Public, statisticians, media	Casual / sophisticated (read only)	**Reads:** Rankings, records, person and competition pages, bulk export. **Writes:** Nothing.
Software Team (DBA role)	DBA	**Reads:** Everything, including catalog and logs. **Writes:** Schema, access rights, backups, export jobs.

Table 4: User roles, categories, and permissions

1. A competitor opens the competition page when registration opens and submits a registration with chosen events; the system accepts it or puts it on the waiting list depending on the competitor limit.

2. During the competition the Delegate's score-taking software enters each attempt; best, average, ranking and record flags are recomputed immediately for live display.

3. After the competition the Delegate submits results; the Results Team checks them and posts them, which updates rankings and records for every country at once.

4. A statistician queries national rankings or downloads the export; a journalist searches a person by partial name.

3 Purpose of the database

The database exists so that one authoritative copy of every attempt produces every record and ranking, consistently, for all users at once. The alternative, per-competition score sheets and spreadsheets, is exactly the file-processing approach Elmasri §1.3 and §1.6 criticise, and the comparison below shows why it fails here.

1. **Handling Frequent Competitors (e.g., Max Park: 218 competitions):**
   - **What a DBMS gives:** One `PERSON` row referenced by every result; WCA ID uniqueness enforced.
   - **What spreadsheets or per-competition files would do:** The same person typed differently in each file; records split across spellings.
2. **Global Records and Ranks:**
   - **What a DBMS gives:** Derived by query or trigger from one result set; updated when any result is posted or corrected.
   - **What spreadsheets or per-competition files would do:** Recomputed by hand; a correction in one file never reaches the record list.
3. **Round Rules and Advancement (Art. 9):**
   - **What a DBMS gives:** Declarative constraints and triggers reject invalid states.
   - **What spreadsheets or per-competition files would do:** Rules checked by eye at the scoring table.
4. **High-Concurrency Registration (e.g., WC2025):**
   - **What a DBMS gives:** Concurrency control and atomic transactions; no over-booking.
   - **What spreadsheets or per-competition files would do:** Two organizers accept the 2000th and 2001st person simultaneously.
5. **Data Confidentiality (Reg. 2d):**
   - **What a DBMS gives:** Views and grants: public views omit them.
   - **What spreadsheets or per-competition files would do:** Whole file shared or nothing.
6. **User-Specific Data Slices:**

- **What a DBMS gives:** External views per user (competitor, delegate, public) over one conceptual schema.
- **What spreadsheets or per-competition files would do:** A separate copy per audience, drifting apart.

7. **Schema Changes (v1 to v2, renamed columns, split tables):**

- **What a DBMS gives:** Logical data independence: views keep old applications working.
- **What spreadsheets or per-competition files would do:** Every program that reads the files rewritten.

8. **System Crashes During Live Score Entry:**

- **What a DBMS gives:** Logging, backup and recovery.
- **What spreadsheets or per-competition files would do:** Lost score sheets.

The database turns the WCA's regulations into enforceable constraints and makes the public rankings a function of the stored attempts. This is better than any level of sophisticated, separately maintained lists.

4 Database Requirements

The design has 10 strong entity types, 4 weak entity types and one specialization of PERSON. Kinds: **K** key, **PK** partial key, **C** composite, **MV** multi-valued, **D** derived, **S** simple. Times are integers in centiseconds, following the WCA export encoding.

4.1 Strong entity types

Attribute	Kind	Domain / constraint
Continent_ID	K	`_Africa`, `_Asia`, `_Europe`, `_North America`, `_Oceania`, `_South America`
Name	K	Unique text, e.g. "Asia"
Record_Code	S	{AfR, AsR, ER, NAR, OcR, SAR}, the continental-record label

Table 5: Strong Entity: CONTINENT

Attribute	Kind	Domain / constraint
Country_ID	K	WCA identifier, e.g. `India`, `USA`, `Hong Kong`
ISO2	K	2 upper-case letters, ISO 3166-1 alpha-2 where one exists, e.g. `IN`, `US`
Name	S	Display name, e.g. "United States", "Hong Kong, China"

Table 6: Strong Entity: COUNTRY (two key attributes)

Specialization of PERSON — overlapping ($d = o$), partial:

- **COMPETITOR**, predicate-defined: WCA_ID IS NOT NULL. Extra attribute First_Year (D) = first 4 digits of WCA_ID.
- **DELEGATE**: Delegate_Status ∈ {trainee, junior, full, regional, senior}; WCA_Email (unique, `@worldcubeassociation.org`); Location (text, e.g. "USA (California)"); Appointed_On (date).

- Overlap is real: most Delegates also compete (Reg. 1j). Partial because organizers or staff may be neither.

Attribute	Kind	Domain / constraint
Person_No	K	Surrogate integer; exists from first registration
WCA_ID	S, unique	`^[0-9]{4}[A-Z]{4}[0-9]{2}$`, e.g. `2012PARK03`; NULL until the person's first results are posted; never reused
Name	C	Romanized_Name (required) + Local_Name (optional, native script), shown as "Ming Zheng ( )"
Gender	S	{m, f, o}
Date_of_Birth	S	Date, not in the future; confidential (Reg. 2d)
Email	S	Confidential contact address
Previous_Identities	MV, C	Set of {Old_Name, Old_Country_ID, Valid_Until}; mirrors the export's `sub_id` history
Competition_Count	D	Number of distinct competitions with at least one RESULT
Medals	D, C	{Gold, Silver, Bronze}: positions 1–3 in final rounds with a valid result

Table 7: Strong Entity (Superclass): PERSON

Attribute	Kind	Domain / constraint
Event_ID	K	e.g. `333`, `333oh`, `minx`, `333mbf`
Event_Name	K	e.g. "3x3x3 Cube", "3x3x3 Multi-Blind"
Display_Rank	S	Positive integer, display order
Result_Kind	S	{time, number, multi}: how attempt values are encoded
Status	S	{official, retired}; 17 official in 2026

Table 8: Strong Entity: EVENT (two key attributes)

Attribute	Kind	Domain / constraint
Format_ID	K	{ `1`, `2`, `3`, `5`, `m`, `a` } = Best of 1/2/3/5, Mean of 3, Average of 5
Format_Name	S	Unique text
Expected_Attempts	S	{1, 2, 3, 5}
Sort_By	S	{single, average}
Trimmed_Per_Side	S	1 for Average of 5, else 0

Table 9: Strong Entity: FORMAT

Attribute	Kind	Domain / constraint
Competition_ID	K	Letters and digits only, e.g. `WC2025`, `IndiaNationals2024`
Name, Short_Name	S	Text
Venue	C	Venue_Name, Street_Address, Venue_Details, City
Coordinates	C	Latitude $\in [-90, 90]$, Longitude $\in [-180, 180]$
Date_Range	C	Start_Date $\leq$ End_Date
Registration_Window	C	Opens_At $<$ Closes_At $\leq$ Start_Date
Entry_Fee	C	Amount (integer, lowest currency unit, ≥ 0) + Currency_Code (ISO 4217, e.g. USD, INR)
Competitor_Limit	S	Positive integer or NULL (no limit)
Refund_Policy_Percent	S	0–100
Website	S	URL
Announced_At, Cancelled_At, Results_Posted_At	S	Timestamps, NULL until the event happens
Championship_Types	MV	Subset of {world, continent `_Asia`..., country ISO2..., `greater_china`}
Duration_Days	D	End_Date − Start_Date + 1
Status	D	{cancelled, upcoming, in progress, past} from Cancelled_At, today and Date_Range
Spots_Left	D	Competitor_Limit − number of accepted REGISTRATIONs

Table 10: Strong Entity: COMPETITION

Attribute	Kind	Domain / constraint
Team_ID	K	e.g. `board`, `wrc`, `wcat`, `wst`
Team_Name	S	e.g. ”WCA Regulations Committee”
Kind	S	{board, committee, team}
Contact_Email	S	Address

Table 11: Strong Entity: TEAM (WCA teams, committees and the Board)

Attribute	Kind	Domain / constraint
Region_ID	K	Integer
Region_Name	S	e.g. "Latin America", "USA (California)"
Level	S	{top, sub}; Senior Delegates lead top regions

Table 12: Strong Entity: REGION (Delegate regions)

Attribute	Kind	Domain / constraint
Role_Code	K	{judge, scrambler, runner, score_taker, announcer, delegate}
Description	S	Text
Regulation_Ref	S	e.g. `1e` for judge, `1f` for scrambler, `1g` for score taker

Table 13: Strong Entity: STAFF_ROLE

Attribute	Kind	Domain / constraint
Result_ID	K	Surrogate integer, as in the WCA export (`results.id`); (round, person) is also unique
Best	D	Best ATTEMPT value per 9f12; DNF (-1) if all failed
Average	D	Per the round's format (9f8–9f11); 0 if the format has no average
Position	D	Rank inside the round; ties share a position (9f15)
Single_Record, Average_Record	D	{WR, AfR, AsR, ER, NAR, OcR, SAR, NR, NULL}, assigned when the result was posted

Table 14: Strong Entity: RESULT (one competitor in one round)

Continued on the next page for Weak entity types...

4.2 Weak entity types

Attribute	Kind	Domain / constraint
Round_Number	PK	1–4 (9m)
Round_Type	S	{first, second, semi_final, final}
Time_Limit	C	Per_Attempt (cs, default 60000 = 10 min) + Cumulative_Scope (MV: set of rounds sharing a cumulative limit)
Cutoff	C	Cutoff_Attempts $\in \{1, 2\}$+ Cutoff_Value (cs or moves), or NULL
Advancement	C	Adv_Type $\in$ {ranking, percent, attempt_result} + Adv_Level (integer); NULL for the final
Scramble_Set_Count	S	≥ 1
Is_Dual, Is_Head_to_Head	S	Booleans
Last_Day	S	Local calendar date the round ends; dates all its results (9i2)
Competitor_Count	D	Number of RESULTs in the round

Table 15: Weak Entity: ROUND (owners: COMPETITION and EVENT)

Attribute	Kind	Domain / constraint
Registrant_No	PK	1, 2, 3, ... per competition
Status	S	{pending, accepted, waiting_list, cancelled, rejected}
Registered_At	S	Timestamp inside the Registration_Window
Guests	S	Integer ≥ 0
Guardian_Consent	S	Boolean; must be true if the person is under 18 on Start_Date (Reg. 2b)

Table 16: Weak Entity: REGISTRATION (owner: COMPETITION)

Attribute	Kind	Domain / constraint
Group_ID	PK part	Letter A, B, C, ...
Is_Extra	PK part	Boolean: spare scramble for extra attempts
Scramble_No	PK part	1–5 ($\leq$ the format's attempts for normal scrambles)
Sequences	MV	One move sequence; several for `333mbf` (one per cube)

Table 17: Weak Entity: SCRAMBLE (owner: ROUND)

Attribute	Kind	Domain / constraint
Attempt_No	PK	1 to the format's Expected_Attempts
Value	S	Integer: -2 DNS, -1 DNF, > 0 encoded result (cs, moves, or Multi-Blind code)
MBLD_Decoded	D, C	{Solved, Attempted, Seconds}, decoded from Value for `333mbf` only

Table 18: Weak Entity: ATTEMPT (owner: RESULT)

4.3 Relationship types

The schema has 23 relationship types: 5 identifying, 1 ternary (ASSIGNED_AS) and 2 recursive (SUPERVISES, SUBREGION_OF). Participation is written **T** (total, every entity must take part) or **P** (partial), in the order of the participants listed.

#	Relationship	Participants	Ratio	Participation	Attributes	Meaning and rule
1	IN_CONTINENT	COUNTRY — CONTINENT	N:1	T, T	—	Every country lies in exactly one continent; every continent has countries
2	REPRESENTS	PERSON — COUNTRY	N:1	T, P	Since_Date	Current country of citizenship represented (Reg. 2e)
3	HOSTED_IN	COMPETITION — COUNTRY	N:1	T, P	—	One host country per competition (multi-location FMC is out of scope)
4	HAS_ROUND (id.)	COMPETITION — ROUND	1:N	T, T	—	A competition holds ≥ 1 round; each round belongs to one competition
5	FOR_EVENT (id.)	EVENT — ROUND	1:N	P, T	—	Each round is of one event; at most 4 rounds per (competition, event) (9m)
6	MAIN_EVENT	COMPETITION — EVENT	N:1	P, P	—	Optional headline event, e.g. `333` for WC2025
7	USES_FORMAT	ROUND — FORMAT	N:1	T, P	—	Each round has exactly one format
8	ALLOWS	EVENT — FORMAT	M:N	T, T	Usage $\in$ {full_round, cut-off_phase}	The format table of Regulation 9b
9	RECEIVES (id.)	COMPETITION — REGISTRATION	1:N	P, T	—	Registrations exist only inside a competition
10	SUBMITS	PERSON — REGISTRATION	1:N	P, T	—	At most one registration per person per competition
11	SELECTS	REGISTRATION — EVENT	M:N	T, P	—	Chosen events; must be events the competition holds

#	Relationship	Participants	Ratio	Participation	Attributes	Meaning and rule
12	ACHIEVES	COMPETITOR — RESULT	1:N	T, T	—	Every result belongs to one competitor; a competitor has ≥ 1 result by definition
13	IN_ROUND	ROUND — RESULT	1:N	P, T	—	Rounds not yet held have no results; at most one result per (round, person)
14	REPRESENTED_AT	RESULT — COUNTRY	N:1	T, P	—	Country the person represented on that day; kept so old national records stay correct after a country change
15	CONSISTS_OF (id.)	RESULT — ATTEMPT	1:N	T, T	—	1 to 5 attempts per result, never more than the format allows
16	SCRAMBLED_FOR (id.)	ROUND — SCRAMBLE	1:N	P, T	—	Scrambles per group of a round
17	DELEGATES	DELEGATE — COMPETITION	M:N	P, T	—	Each competition has ≥ 1 WCA Delegate (Reg. 1a)
18	ORGANIZES	PERSON — COMPETITION	M:N	P, T	—	Each competition has ≥ 1 organizer (Reg. 1a)
19	ASSIGNED_AS (ternary)	PERSON — ROUND — STAFF_ROLE	M:N:P	P, P, P	Slots (MV, composite: {Group_ID, Station_No})	A person serves a round in a role, e.g. judge in group B at station 6
20	SUPERVISES (rec.)	DELEGATE (sup.) — DELEGATE (sup.)	1:N	P, P	—	Senior and Regional Delegates supervise Delegates; every non-senior Delegate has exactly one supervisor
21	SERVES_IN	DELEGATE — REGION	N:1	T, P	—	A Delegate belongs to one region
22	SUBREGION_OF (rec.)	REGION (sub) — REGION (parent)	N:1	P, P	—	e.g. "USA (California)" inside a top-level region
23	MEMBER_OF	PERSON — TEAM	M:N	P, T	Role $\in$ {leader, senior_member, member}, Start_Date, End_Date	Team, committee and Board membership

4.4 Integrity constraints, enforcement and redundancy

Every regulation that can be checked mechanically becomes a constraint with a named enforcement mechanism; the rest stays with the Delegate's judgment (Article 11).

#	Constraint	Source	Enforcement
C1	Keys: Competition_ID, Event_ID, Event_Name, Country_ID, ISO2, WCA_ID (when not NULL), Result_ID are unique; weak entities unique on owner key + partial key	Model	PRIMARY KEY / UNIQUE
C2	Every foreign reference points to an existing row (e.g. RESULT → ROUND, PERSON, COUNTRY)	Ref. integrity	FOREIGN KEY with ON DELETE RESTRICT by default, CASCADE only where stated in 3.3
C3	WCA_ID matches `^[0-9]{4}[A-Z]{4}[0-9]{2}$` and year ≤ year of first competition	Site convention	CHECK + trigger on first result
C4	Gender ∈ {m, f, o}; Delegate_Status, Registration.Status, Round_Type, Adv_Type in sets	Model	CHECK / ENUM
C5	Start_Date ≤ End_Date; Opens_At < Closes_At ≤ Start_Date; Coordinates in range	Model	CHECK
C6	Each competition has ≥ 1 Delegate and ≥ 1 organizer	Reg. 1a	Deferred constraint checked when announced
C7	Each event at most once per competition; at most 4 rounds per event; Round_Number contiguous from 1	Reg. 9j, 9m	Key on (Comp_ID, Event_ID, Round_No) + CHECK 1–4 + trigger
C8	Number of rounds after a round bounded by size: ≤ 99 → 2, ≤ 15 → 1, ≤ 7 → 0	Reg. 9m1–9m3	Trigger when results are finalised
C9	Round format must be allowed for the event (ALLOWS)	Reg. 9b	FK on (Event_ID, Format_ID) into ALLOWS
C10	ATTEMPTs ≤ format's Expected_Attempts; attempts beyond cutoff phase only if a cutoff attempt beats Cutoff_Value	Reg. 9g	Trigger on ATTEMPT insert

#	Constraint	Source	Enforcement
C11	ATTEMPT.Value ≥ -2 and $\neq 0$; time values below 10 min stored to centisecond, 10 min+ to seconds; no value reaches time limit	Reg. 9f1–9f2, A1a4	CHECK + trigger reading ROUND.Time_Limit
C12	A person has RESULT in round $r > 1$ only if non-DNF in round $r-1$ and met advancement; $\geq 25\%$ eliminated unless Dual	Reg. 9p1, 9p2, 9p4, 9v	Trigger on RESULT insert; batch check on close
C13	A person has RESULT in competition only if accepted REGISTRATION selected that event	Reg. 2c1	Trigger on RESULT insert
C14	Accepted registrations $\leq$ Competitor_Limit; extra to waiting_list	Comp. setting	Trigger inside registration transaction (serialisable)
C15	SELECTS only events the competition holds	Model	FK on (Comp_ID, Event_ID) into competition's events
C16	Country change at most once per 12 months unless citizenship lost/acquired	Reg. 2e2	Trigger on REPRESENTS update with override
C17	Head to Head only in final round of `333`, `444`, `333bf`, `333oh`	Reg. 9b7	Trigger on ROUND joined with EVENT
C18	Dual Rounds are rounds 1–2, share format, cutoff, time limit, never final of championship	Reg. 9v1–9v3	Trigger on ROUND
C19	Results can be posted only after End_Date	Reg. 9u	CHECK Results_Posted_At > End_Date
C20	Date_of_Birth and Email readable only by person, organizers, Delegates, WCA staff	Reg. 2d	Views and GRANTs; omitted from public views/export

4.5 Redundancy check

Each logical fact is stored once; the few deliberate copies are controlled and named.

- **Best, Average, Position and record markers on RESULT** are derivable from ATTEMPT rows. We store them (as the WCA export does) because rankings scan millions of results. This is controlled redundancy (Elmasri §1.6.1): a trigger on ATTEMPT recomputes them, and users cannot write them directly.
- **Rankings** (world, continental, national) are not stored at all; they are views over RESULT, so they can never disagree with the results.
- **RESULT → COUNTRY** looks like it duplicates PERSON → COUNTRY but records a different fact: the country on the day of the result. After a country change, old national records must stay with the old country.

- **COMPETITION.Spots_Left, Duration_Days, Status** and **PERSON.Competition_Count, Medals** are derived on read and never stored.
- **Venue and city** live only on COMPETITION; country and continent come through HOSTED_IN and IN_CONTINENT, never copied as text.
- We avoided a REGISTERS_FOR(person, competition, event) ternary: registration status would have repeated once per selected event.

4.6 Bonus items and requirement checklist

All three bonus items are covered: COUNTRY and RESULT illustrate candidate, super and alternate keys; ASSIGNED_AS is rewritten as a weak entity with three binary relationships; SUPERVISES and SUBREGION_OF are recursive.

4.6.1 Candidate, super and alternate keys

COUNTRY(Country_ID, ISO2, Name). Country_ID ("India") and ISO2 ("IN") are each unique on their own.

- Super keys: every attribute set whose values are unique across rows, e.g. {Country_ID}, {ISO2}, {Country_ID, ISO2}, {Country_ID, Name}, {ISO2, Name}, {Country_ID, ISO2, Name}.
- Candidate keys: the minimal super keys, {Country_ID} and {ISO2}. {Country_ID, Name} is a super key but not a candidate key, because dropping Name leaves {Country_ID}, which is still unique.
- Primary key: Country_ID, because every WCA table (results, persons, records) already references countries by it.
- Alternate key: ISO2, the candidate key not chosen as primary; declared UNIQUE NOT NULL.

RESULT gives a second example. Candidate keys are {Result_ID} and {Competition_ID, Event_ID, Round_Number, Person_No}; the second is minimal because one person has one result per round, and removing any of the four attributes allows duplicates. Primary key: Result_ID (short, stable when a round is renumbered). Alternate key: the four-attribute combination.

4.6.2 Rewriting the ternary ASSIGNED_AS

The only relationship of degree > 2 is ASSIGNED_AS(PERSON, ROUND, STAFF_ROLE). It can be replaced by a weak entity and three binary identifying relationships:

- Weak entity **ASSIGNMENT**, partial key Group_ID, attribute Station_No.
- **ASSIGNMENT_OF**: PERSON 1:N ASSIGNMENT (ASSIGNMENT total).
- **ASSIGNMENT_IN**: ROUND 1:N ASSIGNMENT (ASSIGNMENT total).
- **ASSIGNMENT_AS**: STAFF_ROLE 1:N ASSIGNMENT (ASSIGNMENT total).
- Key of ASSIGNMENT: (Person_No, Competition_ID, Event_ID, Round_Number, Role_Code, Group_ID).

This version is better here: a person can judge groups A and C of the same round, and Group_ID as partial key stores that directly instead of as a multi-valued relationship attribute. Three plain binary relationships, PERSON–ROUND, PERSON–STAFF_ROLE and ROUND–STAFF_ROLE, would not be equivalent. From "Asha worked in 333-r1 and 444-r1", "Asha was a judge and a scrambler", and "333-r1 had judges and scramblers", we cannot tell whether Asha judged 333-r1 or scrambled it (Elmasri §3.9).

4.6.3 Same entity type in distinct roles

- **SUPERVISES**(DELEGATE as supervisor, DELEGATE as supervisee), 1:N. A Senior or Regional Delegate supervises many Delegates; each non-senior Delegate has exactly one supervisor. Constraint: the supervisor's status is regional or senior, and the chain has no cycles (trigger).
- **SUBREGION_OF**(REGION as subregion, REGION as parent), N:1, e.g. "USA (California)" under its parent region.

4.6.4 Requirement checklist

Requirement	Minimum	Where it is met
Entity types	5	10 strong + 4 weak + 2 subclasses (2.1)
Entity with two key attributes	1	COUNTRY (Country_ID, ISO2), EVENT (Event_ID, Event_Name), CONTINENT
Weak entity types	2	ROUND, REGISTRATION, SCRAMBLE, ATTEMPT
Relationship types with cardinality and participation	8	23 (2.2)
Relationship of degree ≥ 3	1	ASSIGNED_AS
Subclass	1	COMPETITOR and DELEGATE under PERSON
Composite attributes	2	PERSON.Name, COMPETITION.Venue, Date_Range, Entry_Fee, ROUND.Time_Limit, Cutoff, Advancement
Multi-valued attributes	2	COMPETITION.Championship_Types, PERSON.Previous_Identities, SCRAMBLE.Sequences, ROUND.Time_Limit.Cumulative_Scope
Derived attributes	2	RESULT.Best, Average, Position; COMPETITION.Duration_Days, Spots_Left; PERSON.Competition_Count, Medals
Bonus: candidate / super / alternate key	—	COUNTRY, RESULT
Bonus: n-ary as binaries + weak entity	—	ASSIGNED_AS → ASSIGNMENT
Bonus: same entity in distinct roles	—	SUPERVISES, SUBREGION_OF

Table 21: Requirement checklist and satisfaction mapping

5 Functional requirements

Writing the applications below changed the data requirements in four places; each change is already reflected in section 2.

Functional requirement	**Change it forced in the data requirements**
National rankings and records must stay correct after a competitor changes country (J2, U2)	Added REPRESENTED_AT (RESULT → COUNTRY), separate from PERSON's current country
Staffing-balance report (J4)	Added STAFF_ROLE and the ternary ASSIGNED_AS
Cutoff enforcement on attempt entry (I2)	Made ROUND.Cutoff a composite {Cutoff_Attempts, Cutoff_Value} instead of free text
Waiting-list handling at registration (I1)	Added the waiting_list value to REGISTRATION.Status and the derived Spots_Left

5.1 Retrieval operations

5.1.1 Selection (whole rows matching a condition)

- S1. Retrieve all data of every competition hosted in India whose Start_Date falls in 2026.
- S2. Retrieve all RESULT rows of round 333-r1 of competition WC2025.
- S3. Retrieve all Delegates whose Delegate_Status is regional or senior.
- S4. Retrieve all REGISTRATIONs of a competition whose Status is waiting_list, oldest first.

5.1.2 Projection (chosen attributes only)

- P1. Display the WCA_ID and Name of every competitor currently representing India.
- P2. Display Name, City and Start_Date of every competition that is not cancelled.
- P3. Display Event_ID and Event_Name of all official events in display order.
- P4. Display, for one competition, each round's Event_ID, Round_Number, Format_ID, time limit and cutoff (the round summary printed on the competition's Events tab).

5.1.3 Aggregate

- A1. Minimum Best of all valid (Best > 0) `333` results ever: the 3x3x3 single world record.
- A2. Number of competitions per country per year.
- A3. Average number of competitors per competition, by continent, for the last 12 months.
- A4. Maximum Competitor_Limit over all competitions, and the competition holding it.
- A5. Total attempts recorded per event, and the share of them that are DNF.

5.1.4 Search (partial matching)

- Q1. Persons whose Romanized_Name or Local_Name contains a typed string, e.g. “park” matches ”Max Park”; case-insensitive.
- Q2. Competitions whose Name or City contains “Hyderabad”.
- Q3. Persons whose WCA_ID starts with “2012”, i.e. who first competed in 2012.
- Q4. Venue search: competitions whose Venue_Name contains “Convention”.

5.2 Analysis reports

Each report joins at least three entity types; the join path is given so the query can be written directly.

ID	Report	Entities joined	Output
J1	**Ranking table for an event.** For each competitor, best valid single in event, with world, continental and national rank (ties share rank, Reg. 9f15).	RESULT ⋈ ROUND (event) ⋈ PERSON ⋈ COUNTRY (current) ⋈ CONTINENT	WCA_ID, Name, Country, Best, World / Continental / National rank
J2	**Record history.** Every result that set a world record in an event, oldest first, with country represented on that day.	RESULT ⋈ PERSON ⋈ COUNTRY (via REPRESENTED_AT) ⋈ ROUND ⋈ COMPETITION	Date (Last_Day), Name, Country, Value, Competition
J3	**Medal table of World Championships.** Gold, silver, bronze per country across final rounds of competitions with world championship type.	RESULT ⋈ ROUND (Final) ⋈ COMPETITION ⋈ COUNTRY (via REPRESENTED_AT)	Country, Gold, Silver, Bronze, sorted by medals
J4	**Staffing balance.** Per competition, competitors who competed in more rounds than assigned as judge/scrambler (Reg. 1e2, 1f2).	PERSON ⋈ RESULT ⋈ ROUND ⋈ ASSIGNED_AS ⋈ STAFF_ROLE	Name, rounds competed, staff assignments, difference
J5	**Delegate workload.** Delegates who delegated more competitions in last 12 months than region average, with supervisor name.	DELEGATE ⋈ DELEGATES ⋈ COMPETITION ⋈ REGION ⋈ SUPERVISES	Delegate, Region, count, regional average, Supervisor
J6	**Cutoff pass rate.** For each event, over 2026 competitions, share of competitors in cutoff rounds who earned remaining attempts.	ROUND ⋈ RESULT ⋈ ATTEMPT ⋈ EVENT ⋈ COMPETITION	Event, competitors in cutoff rounds, passed, pass %

ID	Report	Entities joined	Output
J7	**Newcomer retention by country.** Of persons whose first competition was in year Y, share who competed again within 12 months.	PERSON ⋈ RESULT ⋈ ROUND ⋈ COMPETITION ⋈ COUNTRY	Country, newcomers, returning within 12 months, %

J5 is the WCA analogue of the assignment's employee-versus-supervisor example: it compares each Delegate with an aggregate over their region and reports their supervisor through the recursive SUPERVISES.

5.3 Modification operations

Each operation runs as one transaction: if any check fails, nothing is written. Constraint numbers refer to section 2.3.

5.3.1 Insertions with integrity checks

I1. Register a person for a competition.

1. The competition exists, is not cancelled, and the current time is inside its Registration_Window.
2. The person has no other REGISTRATION for this competition (key on (Competition_ID, Person_No)).
3. At least one event is selected, and every selected event is held at this competition (C15).
4. If the person is under 18 on Start_Date, Guardian_Consent must be true (Reg. 2b).
5. Registrant_No = (largest Registrant_No in the competition) + 1.
6. Status = accepted if Spots_Left > 0, otherwise waiting_list (C14). Steps 5–6 run under serialisable isolation, so two simultaneous registrations cannot both take the last place.

I2. Enter one attempt during a round.

1. The person holds an accepted registration that selected the round's event (C13).
2. For round 2 onward, the person met the previous round's advancement condition (C12).
3. Attempt_No $\leq$ format's Expected_Attempts, and that number is not already used for this result (C10, key).
4. If the round has a cutoff and Attempt_No $>$ Cutoff_Attempts, at least one earlier attempt beat Cutoff_Value (Reg. 9g).
5. The value is in its domain and below the per-attempt and cumulative time limits; otherwise it must be entered as DNF (C11).
6. On success, the trigger recomputes RESULT.Best, Average and the Position of every result in the round.

I3. Announce a new competition. Competition_ID is new and alphanumeric; dates and registration window are consistent (C5); the host country exists; at least one Delegate and one organizer are linked (C6); every round's format is allowed for its event (C9) and round numbers per event are 1, 2, ... with at most 4 (C7). Announced_At is then set.

5.3.2 Updates with cascading business rules

U1. Correct a wrongly entered attempt (score-taking mistake found later, Reg. 1g2 and 9t).

1. Update ATTEMPT.Value.
2. Recompute the RESULT's Best and Average, then every Position in that round.
3. Re-evaluate record markers for that event from that date onward, so a revoked record passes to the next valid result.
4. If the competitor no longer qualifies for the next round, delete their results in all later rounds of that event (9t). Other competitors' results are not touched, even if one of them should have advanced instead (9t1, 9t3).
5. Rankings update on their own, because they are views.

U2. Change the country a person represents (Reg. 2c2, 2e2).

1. Reject if the last change was under 12 months ago, unless the Results Team sets the citizenship-lost or citizenship-acquired override.
2. Append {current name, current country, today} to Previous_Identities.
3. Update REPRESENTS. Existing RESULT → COUNTRY links stay unchanged, so old national records remain with the old country, while J1 ranks the person under the new one.

U3. Cancel a competition.

1. Reject if any RESULT exists for it.
2. Set Cancelled_At = now.
3. Set every REGISTRATION Status to cancelled; delete its ASSIGNED_AS rows and unused SCRAMBLEs.
4. Status becomes cancelled automatically; the competition stays visible for history.

U4. Change a Delegate's status. When a Regional Delegate steps down, each of their supervisees is re-linked to the Senior Delegate of the parent region (SUPERVISES). If a Delegate becomes a trainee, any upcoming competition where they are the only Delegate is flagged, because a trainee cannot delegate alone.

U5. Promote from the waiting list. When an accepted registration is cancelled before Start_Date, the earliest waiting_list registration (by Registered_At) becomes accepted, provided Spots_Left > 0.

5.3.3 (c) Deletion with referential integrity

D1. Delete a withdrawn registration.

1. Allowed only before Start_Date and only if the person has no RESULT in this competition (ON DELETE RESTRICT from RESULT).
2. Cascades: its SELECTS rows and the person's ASSIGNED_AS rows in this competition's rounds are deleted.
3. Triggers U5 for the freed place.

D2. Delete a draft (never announced) competition. Rejected if any RESULT references it. Otherwise ROUND, SCRAMBLE, REGISTRATION, SELECTS, ASSIGNED_AS, DELEGATES and ORGANIZES rows are deleted with it (ON DELETE CASCADE). PERSON, EVENT and COUNTRY rows are never deleted by this cascade.

6 Assumptions, open points and sources

Most facts above were read from the WCA site, accessed in October 2026; the items below are either our modelling choices or details we could not confirm from a page.

6.1 Assumptions and interpretations

- "Functional requirements must make at least one change to the data requirements": read as "the applications must have influenced the schema"; the table at the top of section 3 lists four such changes. If the assignment meant "must include modifications", section 3.3 covers that too.
- PERSON uses a surrogate key because newcomers register before they receive a WCA ID, and early competitors have WCA IDs without website accounts.
- Competition names are not assumed unique; only Competition_ID is a key.
- SUBREGION_OF is inferred from Delegate location strings and the Delegate Handbook; we did not find a published region table.
- Medals are counted as positions 1–3 in final rounds with a valid result; this matches the person page totals in spirit, but we did not verify the WCA's exact rule.

LLM Usage

LLMs were utilised for data retrieval, formatting, and typesetting. Quick searches over of the website were made with their assistance.

6.2 Sources

- WCA Regulations, version April 1, 2026
- WCA Results Export and README, export format 2.0.2
- Competition API: WC2025
- Public WCIF: WC2025
- Person API: 2012PARK03
- Speedcubing History
- WCA Delegate Handbook
- Elmasri & Navathe, *Fundamentals of Database Systems*, 7th ed., Ch. 1–4